

PERSONAL LEARNING LOG

Week 1: Python Fundamentals and Data Handling

- Activities:
 - Learned Python programming concepts
 - Worked with NumPy and Pandas
 - Loaded and explored telecom KPI datasets
 - Performed basic data analysis
- Learning Outcomes:
 - Data manipulation using Pandas
 - Reading and writing CSV files
 - Understanding telecom KPI datasets

Week 2: Network Health Classification using Telecom KPI data

- Activities:
 - Explored and analyzed the telecom KPI dataset using EDA
 - Created visualizations to understand network performance
 - Performed feature engineering and data preprocessing
 - Built and trained Logistic Regression, Decision Tree, and Random Forest models
 - Applied cross-validation and hyperparameter tuning to improve model performance
 - Evaluated models using classification metrics, confusion matrices, and ROC curves
 - Analyzed feature importance and compared model performance
 - Documented the project findings in a technical report
- Learning Outcomes:
 - Learned the complete end-to-end machine learning workflow
 - Gained hands-on experience with data preprocessing and feature engineering
 - Understood how to train, tune, and evaluate classification models
 - Learned to use Scikit-learn for model development and performance analysis
 - Developed the ability to interpret model results and identify the best-performing algorithm for telecom network health classification

Week 3: Telecom Network KPI Anomaly Detection using PyTorch

- Activities:

- Studied neural network and CNN fundamentals
- Preprocessed telecom KPI data and created PyTorch datasets
- Built and trained an MLP model using PyTorch
- Performed learning rate experiments and hyperparameter tuning
- Visualized training performance and evaluated the model using classification metrics
- Implemented a simple CNN on the MNIST dataset
- Learning Outcomes:
 - Understood neural networks, CNNs, and deep learning workflows
 - Learned data preprocessing and model training using PyTorch
 - Gained experience in model evaluation and performance visualization
 - Improved understanding of hyperparameter tuning and overfitting analysis
 - Developed practical skills in implementing MLP and CNN models for classification tasks

Week 4: 5G Architecture and Telecom Fundamentals

- Activities:
 - Studied 5G NR architecture, radio technologies, and core network functions
 - Analyzed telecom KPIs and prepared KPI threshold tables
 - Performed packet capture and protocol analysis using Wireshark
 - Studied TCP, DNS, HTTP, and end-to-end 5G data session flows
 - Conducted a telecom network case study and proposed optimization measures
 - Prepared technical reports, architecture diagrams, and presentation slides
- Learning Outcomes:
 - Gained understanding of 5G network architecture and core concepts
 - Learned the concepts of OFDM, Beamforming, Massive MIMO, Network Slicing, and O-RAN
 - Developed practical skills in network packet analysis and protocol identification
 - Understood telecom KPIs and their impact on network performance
 - Improved troubleshooting and root cause analysis skills for telecom networks
 - Enhanced technical documentation and presentation skills

Week 5: AI Operations (AIOps) - Network Anomaly Detection and Root Cause Analysis

- Activities:
 - Explored and visualized telecom KPI time-series data

- Developed anomaly detection models using Isolation Forest and LSTM
- Performed threshold tuning for anomaly detection
- Forecasted future network traffic using time-series models
- Conducted Root Cause Analysis (RCA) and generated recommendations
- Built a QoS/QoE prediction model and designed KPI labeling strategies
- Integrated AI models into an AIOps workflow and documented project findings
- Learning Outcomes:
 - Gained practical experience in telecom network analytics and AIOps
 - Learned time-series analysis, anomaly detection, forecasting, and machine learning techniques
 - Improved skills in model evaluation, KPI interpretation, and root cause analysis
 - Enhanced understanding of predictive maintenance, QoS prediction, and AI-driven network monitoring
 - Strengthened proficiency in Python, data visualization, and end-to-end AI project development

Week 6: dashboard Development

- Activities:
 - Studied Transformer architecture and prompt engineering
 - Built a telecom RAG knowledge base using LangChain and ChromaDB
 - Developed a PAG pipeline and Streamlit chatbot
 - Compared chunking strategies and embedding models
 - Evaluated chatbot performance and explored LoRA/PEFT
 - Integrated an LLM API and completed project documentation
- Learning Outcomes:
 - Understood Transformer models and prompt engineering techniques
 - Learned to build and optimize a RAG-based chatbot
 - Gained experience with vector databases, embeddings, and LLM integration
 - Improved skills in chatbot evaluation, fine-tuning concepts, and technical documentation

Week 7: AI-Based Telecom Network Solution

- Activities:
 - Developed a complete AI-based telecom KPI anomaly detection pipeline, including data preprocessing, feature engineering, model training, evaluation, hyperparameter tuning, and an interactive Streamlit dashboard

- Implemented the Isolation Forest algorithm for anomaly detection, documented the project with a README and reports, and managed the project using GitHub with proper version control
- Learning Outcomes:
 - Gained practical experience in end-to-end machine learning workflow, including data preparation, unsupervised anomaly detection, model evaluation and dashboard development
 - Improved skills in Python, Scikit-learn, Streamlit, Git/GitHub, project documentation, and presenting AI solutions for real-world telecom network monitoring.

Week 8: Project Finalization

- Activities:
 - Refined and optimized the capstone project by improving code quality, documentation, and repository organization
 - Prepared technical documents, installation guide, evaluation report, and final project presentation
 - Validated the complete AI pipeline, demonstrated the Streamlit application, and finalized the GitHub repository for submission
- Learning Outcomes:
 - Improved skills in code refactoring, technical documentation, and software project organization
 - Gained experience in presenting AI solutions, demonstrating end-to-end workflows, and maintaining a professional GitHub repository
 - Enhanced understanding of software deployment readiness, reproducibility, and project documentation best practices